Hongbo Liu, Ph.D.

Appointment

Assistant Professor, Department of Biomedical Genetics
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Education and Training

• 2018 – 2023	Postdoctoral Fellow (mentor: Dr. Katalin Susztak)	University of Pennsylvania
• 2016 – 2018	Postdoctoral Fellow (mentor: Dr. Hui Shen)	Van Andel Institute
• 2012 – 2015	Ph.D., Biomedical Engineering (advisor: Dr. Qiong Wu)	Harbin Institute of Technology
• 2007 – 2010	MSc, Biomedical Engineering (advisor: Dr. Yan Zhang)	Harbin Medical University
• 2003 – 2007	B.S., Information and Computing Science	Qufu Normal University

Professional Experience

• 2024 – present	Assistant Professor	Department of Biomedical Genetics, University of Rochester
• 2024 – present	Member	University of Rochester Aging Institute
• 2026 – present	Member	Wilmot Cancer Institute
• 2023 – 2024	Research Associate	Department of Medicine, University of Pennsylvania
• 2015 – 2016	Associate Professor	College of Bioinformatics Sci. & Tech., Harbin Medical University
• 2010 – 2012	Research Assistant	College of Bioinformatics Sci. & Tech., Harbin Medical University

Publications

(Total: 83 peer-reviewed publications | * Equal contribution | # Corresponding author | Listed in reverse chronological order | Click on journal names to view publications on their respective websites | Citation metrics on Google Scholar)

1. Dean Y Zhang, Hufeng Zhou, Maya U Sheth, Andreas R Gschwind, Jesse M Engreitz, Xihong Lin, and **Hongbo Liu**[#]. mV2G: a multiomic atlas for tissue-specific variant-to-gene prioritization, *Nucleic Acids Research*, gkag1089. (2026)
2. Shen Li, **Hongbo Liu**, Sujin Lee, Eunji Ha, Bernhard Dumoulin, Matthew B Palmer, Clint L Miller, Scott M Damrauer, Rajeev Malhotra, and Katalin Susztak[#]. Transcription factor TBX2 is a key regulator of vascular calcification, *European Heart Journal*, ehag756. (2026)
3. Yanjuan Hou, Konstantin Kloetzer, Bernhard Dumoulin, Eunji Ha, **Hongbo Liu**, Dhanunjay Mukhi, Magaiver Andrade-Silva, Xuemei Hu, Siyu Pan, and Katalin Susztak[#]. ELF3 links kidney function GWAS loci to maladaptive epithelial inflammation, *Science Advances*, 12(30):eaec6741. (2026)
4. Lu Li, Xinyue Mao, Linda Xiaoyan Li, Huaping Xiao, Zhenkun Lou, **Hongbo Liu**, Julie Xia Zhou, Li Yao, and Xiaogang Li[#]. Feedback loops between DNMT1 and autophagy as well as senescence promotes organ aging and canities, *Autophagy*, 22(10):2809-2830. (2026)
5. Lakshmi P Kolligundla*, Dhanunjay Mukhi*, Jonathan Levinsohn, Andi Bergerson, Yanjuan Hou, **Hongbo Liu**, Brain C Capell, and Katalin Susztak[#]. KDM1A regulates nephron progenitor programs and is associated with human kidney disease risk, *Nephrology Dialysis Transplantation*, 41(11):2027-2041. (2026)
6. **Hongbo Liu**[#], Amin Abedini, Eunji Ha, Ziyuan Ma, Xin Sheng, Bernhard Dumoulin, Chengxiang Qiu, Tamas Aranyi, Shen Li, Nicole Dittrich, Hua-Chang Chen, Ran Tao, Der-Cherng Tarng, Feng-Jen Hsieh, Shih-Ann Chen, Shun-Fa Yang, Mei-Yueh Lee, Pui-Yan Kwok, Jer-Yuarn Wu, Chien-Hsiun Chen, Atlas Khan, Nita A. Limdi, Wei-Qi Wei, Theresa L. Walunas, Elizabeth W. Karlson, Eimear E. Kenny, Yuan Luo, Leah Kottyan, John J. Connolly, Gail P.

Jarvik, Chunhua Weng, Ning Shang, Joanne B. Cole, Josep M. Mercader, Ravi Mandla, Timothy D. Majarian, Jose C. Florez, Mary E. Haas, Luca A. Lotta, Theodore G. Drivas, Ha My T. Vy, Girish N. Nadkarni, Laura K. Wiley, Melissa P. Wilson, Christopher R. Gignoux, Humaira Rasheed, Laurent F. Thomas, Bjørn Olav Åsvold, Ben M. Brumpton, Stein I. Hallan, Kristian Hveem, Jie Zheng, Jacklyn N. Hellwege, Matthew Zawistowski, Sebastian Zöllner, Nora Franceschini, Hailong Hu, Jianfu Zhou, Krzysztof Kiryluk, Marylyn D. Ritchie, Matthew Palmer, Todd L. Edwards, Benjamin F. Voight, Adriana M. Hung and Katalin Susztak[#]. Kidney multiome-based genetic scorecard reveals convergent coding and regulatory variants, *Science*, 387, eadp4753. (2025)

7. Shen Li, **Hongbo Liu**, Hailong Hu, Eunji Ha, Praveena Prasad, Brenita C. Jenkins, Ujjalkumar Subhash Das, Sarmistha Mukherjee, Kyosuke Shishikura, Renming Hu, Daniel J. Rader, Liming Pei, Joseph A. Baur, Megan L. Matthews, Garret A. FitzGerald, Melanie R. McReynolds and Katalin Susztak[#]. Human genetics identify convergent signals in mitochondrial LACTB-mediated lipid metabolism in cardiovascular-kidney-metabolic syndrome, *Cell Metabolism*, 37, 154-168 e157. (2025)
8. Daigoro Hirohama, João Fadista, Eunji Ha, **Hongbo Liu**, Amin Abedini, Jonathan Levinsohn, Allison Vassalotti, Li Zeng, Chenyu Li, Samer Mohandes, Steven Vitale, Dmitry Shungin, Thao Nguyen, Monika A Niewczas, Niclas Olsson, Fiona E McAllister, Anil Karihaloo, and Katalin Susztak[#]. The proteogenomic landscape of the human kidney and implications for cardio-kidney-metabolic health, *Nature Medicine*, 31(11):3917-3929. (2025)
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11. Joanne B Cole[#], Emma H Dahlström, Damian Fermin, Yogesh Gupta, Claire Hill, Laura J Smyth, **Hongbo Liu**, Raymond J Kreienkamp, Marcus G Pezzolesi, Jing Jing Cao, Erkka Valo, Wei-Min Chen, Suna Onengut-Gumuscu, Stephen S Rich, Eoin P Brennan, Darrell Andrews, Ciarán Kennedy, Harvest F Gu, Lars Stechemesser, Raimund Weitgasser, Jelizaveta Sokolovska, Lina Radzeviciene, Rasa Verkauskiene, Nicolae M Panduru, Peter Rossing, Tarunveer S Ahluwalia, Gianpaolo Zerbini, Michel Marre, Samy Hadjadj, Tina Costacou, Rachel G Miller, Barbara E Klein, Kristine E Lee, Janet K Snell-Bergeon, Maria Luiza Caramori, Michael Mauer, Kerstin Brismar, Petter Bjornstad, Amy J McKnight, Gareth McKay, Viji Nair, Rany M Salem, Per-Henrik Groop, Catherine Godson, Katalin Susztak, Matthias Kretzler, Alexander P Maxwell, Andrzej Krolewski, Andrew Paterson, Niina Sandholm-Lafferre, Jose C Florez, and Joel N Hirschhorn[#]. Genome-wide association study of quantitative kidney function in 52,531 individuals with diabetes identifies five diabetes-specific loci, *Journal of the American Society of Nephrology*, 36(10):1939-1953. (2025)
12. Xiujie Liang, **Hongbo Liu**, Hailong Hu, Eunji Ha, Jianfu Zhou, Amin Abedini, Andrea Sanchez-Navarro, Konstantin A. Klotzer and Katalin Susztak[#]. TET2 germline variants promote kidney disease by impairing DNA repair and activating cytosolic nucleotide sensors, *Nature Communications*, 15, 9621. (2024)
13. Teresa K. Chen[#], Aditya L. Surapaneni, Insa M. Schmidt, Sushrut S. Waikar, Josef Coresh, **Hongbo Liu**, Katalin Susztak, Eugene P. Rhee, Celina Liu, Pascal Schlosser and Morgan E. Grams. Proteomics and Incident Kidney Failure in Individuals With CKD: The African American Study of Kidney Disease and Hypertension and the Boston Kidney Biopsy Cohort, *Kidney Medicine*, 6, 100921. (2024)
14. Tanya J. Major*, Riku Takei*, Hirotaka Matsuo*, Megan P. Leask*, Nicholas A. Sumpter*, Ruth K. Topless*, Yuya Shirai*, Wei Wang*, Murray J. Cadzow, Amanda J. Phipps-Green, Zhiqiang Li, Aichang Ji, Marilyn E. Merriman, Emily Morice, Eric E. Kelley, Wen-Hua Wei, Sally P. A. McCormick, Matthew J. Bixley, Richard J. Reynolds, Kenneth G. Saag, Tayaza Fadason, Evgenia Golovina, Justin M. O'Sullivan, Lisa K. Stamp, Nicola Dalbeth, Abhishek Abhishek, Michael Doherty, Edward Roddy, Lennart T. H. Jacobsson, Meliha C. Kapetanovic, Olle Melander, Mariano Andres, Fernando Perez-Ruiz, Rosa J. Torres, Timothy Radstake, Timothy L. Jansen, Matthijs Janssen, Leo A. B. Joosten, Ruiqi Liu, Orsolya I. Gaal, Tania O. Crisan, Simona Rednic, Fina Kurreeman, Tom W. J. Huizinga, Rene Toes, Frederic Liote, Pascal Richette, Thomas Bardin, Hang Korng Ea, Tristan Pascart, Geraldine M. McCarthy, Laura Helbert, Blanka Stiburkova, Anne- K. Tausche, Till Uhlig, Veronique Vitart, Thibaud S. Boutin, Caroline Hayward, Philip L. Riches, Stuart H. Ralston, Archie Campbell, Thomas M. MacDonald, Akiyoshi Nakayama, Tappei Takada, Masahiro Nakatochi, Seiko Shimizu, Yusuke Kawamura, Yu Toyoda, Hirofumi Nakaoka, Ken Yamamoto, Keitaro Matsuo, Nariyoshi Shinomiya, Kimiyoshi Ichida, Chaeyoung Lee, Linda A. Bradbury, Matthew A. Brown, Philip C. Robinson, Russell R. C. Buchanan, Catherine L. Hill, Susan Lester, Malcolm D. Smith, Maureen Rischmueller, Hyon K. Choi, Eli A. Stahl, Jeff N. Miner, Daniel H. Solomon, Jing Cui, Kathleen M. Giacomini, Deanna J. Brackman, Eric M. Jorgenson, **Hongbo Liu**, Katalin Susztak, Suyash Shringarpure, Alexander So, Yukinori Okada, Changgui Li,

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15. Amin Abedini*, Jonathan Levinsohn*, Konstantin A. Klotzer, Bernhard Dumoulin, Ziyuan Ma, Julia Frederick, Poonam Dhillon, Michael S. Balzer, Rojesh Shrestha, **Hongbo Liu**, Steven Vitale, Andi M. Bergeson, Kishor Devalaraja-Narashimha, Paola Grandi, Tanmoy Bhattacharyya, Erding Hu, Steven S. Pullen, Carine M. Boustany-Kari, Paolo Guarnieri, Anil Karihaloo, Daniel Traum, Hanying Yan, Kyle Coleman, Matthew Palmer, Lea Sarov-Blat, Lori Morton, Christopher A. Hunter, Klaus H. Kaestner, Mingyao Li and Katalin Susztak[#]. Single-cell multi-omic and spatial profiling of human kidneys implicates the fibrotic microenvironment in kidney disease progression, *Nature Genetics*, 56, 1712-1724. (2024)
16. Edward D. Siew*[#], Jacklyn N. Hellwege*, Adriana M. Hung, Bethany C. Birkelo, Andrew J. Vincz, Sharidan K. Parr, Jason Denton, Robert A. Greevy, Cassianne Robinson-Cohen, **Hongbo Liu**, Katalin Susztak, Michael E. Matheny and Digna R. Velez Edwards. Genome-wide association study of hospitalized patients and acute kidney injury, *Kidney International*, 106, 291-301. (2024)
17. Abel Fothi, **Hongbo Liu**, Katalin Susztak and Tamas Aranyi[#]. Improve-RRBS: a novel tool to correct the 3' trimming of reduced representation sequencing reads, *Bioinformatics Advances*, 4, vbae076. (2024)
18. Yu Yan, **Hongbo Liu**, Amin Abedini, Xin Sheng, Matthew Palmer, Hongzhe Li and Katalin Susztak[#]. Unraveling the epigenetic code: human kidney DNA methylation and chromatin dynamics in renal disease development, *Nature Communications*, 15, 873. (2024)
19. Amin Abedini, Andrea Sanchez-Navaro, Junnan Wu, Konstantin A. Klotzer, Ziyuan Ma, Bibek Poudel, Tomohito Doke, Michael S. Balzer, Julia Frederick, Hana Cernecka, **Hongbo Liu**, Xiujie Liang, Steven Vitale, Peter Kolkhof and Katalin Susztak[#]. Single-cell transcriptomics and chromatin accessibility profiling elucidate the kidney-protective mechanism of mineralocorticoid receptor antagonists, *The Journal of Clinical Investigation*, 134, (2024)
20. Dhanunjay Mukhi, Lingzhi Li, **Hongbo Liu**, Tomohito Doke, Lakshmi P. Kolligundla, Eunji Ha, Konstantin Kloetzer, Amin Abedini, Sarmistha Mukherjee, Junnan Wu, Poonam Dhillon, Hailong Hu, Dongyin Guan, Katsuhiko Funai, Kahealani Uehara, Paul M. Titchenell, Joseph A. Baur, Kathryn E. Wellen and Katalin Susztak[#]. ACSS2 gene variants determine kidney disease risk by controlling de novo lipogenesis in kidney tubules, *The Journal of Clinical Investigation*, 134(4):e172963, (2024)
21. Jianfu Zhou, Amin Abedini, Michael S. Balzer, Rojesh Shrestha, Poonam Dhillon, **Hongbo Liu**, Hailong Hu and Katalin Susztak[#]. Unified Mouse and Human Kidney Single-Cell Expression Atlas Reveal Commonalities and Differences in Disease States, *Journal of the American Society of Nephrology*, 34, 1843-1862. (2023)
22. Farah Lizotte, Marina Rousseau, Benoit Denhez, Dominique Levesque, Andreanne Guay, **HongBo Liu**, Julie Moreau, Sarah Higgins, Robert Sabbagh, Katalin Susztak, Francois-Michel Boisvert, Anne Marie Cote and Pedro Geraldes[#]. Deletion of protein tyrosine phosphatase SHP-1 restores SUMOylation of podocin and reverses the progression of diabetic kidney disease, *Kidney International*, 104, 787-802. (2023)
23. Pascal Schlosser[#], Jingning Zhang, **Hongbo Liu**, Aditya L. Surapaneni, Eugene P. Rhee, Dan E. Arking, Bing Yu, Eric Boerwinkle, Paul A. Welling, Nilanjan Chatterjee, Katalin Susztak, Josef Coresh and Morgan E. Grams. Transcriptome- and proteome-wide association studies nominate determinants of kidney function and damage, *Genome Biology*, 24, 150. (2023)
24. Kelly Yichen Li, Claudia Ha Ting Tam, **Hongbo Liu**, Samantha Day, Cadmon King Poo Lim, Wing Yee So, Chuiguo Huang, Guozhi Jiang, Mai Shi, Heung Man Lee, Hui-Yao Lan, Cheuk-Chun Szeto, Robert L. Hanson, Robert G. Nelson, Katalin Susztak, Juliana C. N. Chan, Kevin Y. Yip[#] and Ronald C. W. Ma[#]. DNA methylation markers for kidney function and progression of diabetic kidney disease, *Nature Communications*, 14, 2543. (2023)
25. Daigoro Hirohama, Amin Abedini, Salina Moon, Aditya Surapaneni, Simon T. Dillon, Allison Vassalotti, **Hongbo Liu**, Tomohito Doke, Victor Martinez, Zaipul Md Dom, Anil Karihaloo, Matthew B. Palmer, Josef Coresh, Morgan E. Grams, Monika A. Niewczas[#] and Katalin Susztak[#]. Unbiased Human Kidney Tissue Proteomics Identifies Matrix Metalloproteinase 7 as a Kidney Disease Biomarker, *Journal of the American Society of Nephrology*, 34, 1279-1291. (2023)
26. Poonam Dhillon*, Kelly Ann Mulholland*, Hailong Hu, Jihwan Park, Xin Sheng, Amin Abedini, **Hongbo Liu**, Allison Vassalotti, Junnan Wu and Katalin Susztak[#]. Increased levels of endogenous retroviruses trigger fibroinflammation and play a role in kidney disease development, *Nature Communications*, 14, 559. (2023)
27. Xiujie Liang, Tamas Aranyi, Jianfu Zhou, Yuting Guan, Hailong Hu, **Hongbo Liu** and Katalin Susztak[#]. Tet2- and Tet3- Mediated Cytosine Hydroxymethylation in Six2 Progenitor Cells in Mice Is Critical for Nephron Progenitor Differentiation and Nephron Endowment, *Journal of the American Society of Nephrology*, 34, 572-589. (2023)

28. **Hongbo Liu**, Tomohito Doke, Dong Guo, Xin Sheng, Ziyuan Ma, Joseph Park, Ha My T. Vy, Girish N. Nadkarni, Amin Abedini, Zhen Miao, Matthew Palmer, Benjamin F. Voight, Hongzhe Li, Christopher D. Brown, Marylyn D. Ritchie, Yan Shu and Katalin Susztak[#]. Epigenomic and transcriptomic analyses define core cell types, genes and targetable mechanisms for kidney disease, *Nature Genetics*, 54, 950-962. (2022)

29. Niina Sandholm*, Joanne B. Cole*, Viji Nair, Xin Sheng, **Hongbo Liu**, Emma Ahlqvist, Natalie van Zuydam, Emma H. Dahlstrom, Damian Fermin, Laura J. Smyth, Rany M. Salem, Carol Forsblom, Erkka Valo, Valma Harjutsalo, Eoin P. Brennan, Gareth J. McKay, Darrell Andrews, Ross Doyle, Helen C. Looker, Robert G. Nelson, Colin Palmer, Amy Jayne McKnight, Catherine Godson, Alexander P. Maxwell, Leif Groop, Mark I. McCarthy, Matthias Kretzler, Katalin Susztak, Joel N. Hirschhorn[#], Jose C. Florez and Per-Henrik Groop[#]. Genome-wide meta-analysis and omics integration identifies novel genes associated with diabetic kidney disease, *Diabetologia*, 65, 1495-1509. (2022)

30. Xin Sheng*, Yuting Guan*, Ziyuan Ma, Junnan Wu, **Hongbo Liu**, Chengxiang Qiu, Steven Vitale, Zhen Miao, Matthew J. Seasock, Matthew Palmer, Myung K. Shin, Kevin L. Duffin, Steven S. Pullen, Todd L. Edwards, Jacklyn N. Hellwege, Adriana M. Hung, Mingyao Li, Benjamin F. Voight, Thomas M. Coffman, Christopher D. Brown and Katalin Susztak[#]. Mapping the genetic architecture of human traits to cell types in the kidney identifies mechanisms of disease and potential treatments, *Nature Genetics*, 53, 1322-1333. (2021)

31. Tomohito Doke, Shizheng Huang, Chengxiang Qiu, **Hongbo Liu**, Yuting Guan, Hailong Hu, Ziyuan Ma, Junnan Wu, Zhen Miao, Xin Sheng, Jianfu Zhou, Aili Cao, Jianhua Li, Lewis Kaufman, Adriana Hung, Christopher D. Brown, Richard Pestell and Katalin Susztak[#]. Transcriptome-wide association analysis identifies DACH1 as a kidney disease risk gene that contributes to fibrosis, *The Journal of Clinical Investigation*, 131(10):e141801. (2021)

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33. Zhen Miao*, Michael S. Balzer*, Ziyuan Ma, **Hongbo Liu**, Junnan Wu, Rojesh Shrestha, Tamas Aranyi, Amy Kwan, Ayano Kondo, Marco Pontoglio, Junhyong Kim, Mingyao Li, Klaus H. Kaestner and Katalin Susztak[#]. Single cell regulatory landscape of the mouse kidney highlights cellular differentiation programs and disease targets, *Nature Communications*, 12, 2277. (2021)

34. Yuting Guan*, Xiujie Liang*, Ziyuan Ma, Hailong Hu, **Hongbo Liu**, Zhen Miao, Andreas Linkermann, Jacklyn N. Hellwege, Benjamin F. Voight and Katalin Susztak[#]. A single genetic locus controls both expression of DPEP1/CHMP1A and kidney disease development via ferroptosis, *Nature Communications*, 12, 5078. (2021)

35. Pascal Schlosser*[#], Adrienne Tin*, Pamela R. Matias-Garcia, Chris H. L. Thio, Roby Joehanes, **Hongbo Liu**, Antoine Weihs, Zhi Yu, Anselm Hoppmann, Franziska Grundner-Culemann, Josine L. Min, Adebowale A. Adeyemo, Charles Agyemang, Johan Arnlov, Nasir A. Aziz, Andrea Baccarelli, Murielle Bochud, Hermann Brenner, Monique M. B. Breteler, Cristian Carmeli, Layal Chaker, John C. Chambers, Shelley A. Cole, Josef Coresh, Tanguy Corre, Adolfo Correa, Simon R. Cox, Niek de Klein, Graciela E. Delgado, Arce Domingo-Relloso, Kai-Uwe Eckardt, Arif B. Ekici, Karlhans Endlich, Kathryn L. Evans, James S. Floyd, Myriam Fornage, Lude Franke, Eliza Fraszczyk, Xu Gao, Xin Gao, Mohsen Ghanbari, Sahar Ghasemi, Christian Gieger, Philip Greenland, Megan L. Grove, Sarah E. Harris, Gibran Hemani, Peter Henneman, Christian Herder, Steve Horvath, Lifang Hou, Mikko A. Hurme, Shih-Jen Hwang, Marjo-Riitta Jarvelin, Sharon L. R. Kardia, Silva Kasela, Marcus E. Kleber, Wolfgang Koenig, Jaspal S. Kooner, Holly Kramer, Florian Kronenberg, Brigitte Kuhnel, Terho Lehtimaki, Lars Lind, Dan Liu, Yongmei Liu, Donald M. Lloyd-Jones, Kurt Lohman, Stefan Lorkowski, Ake T. Lu, Riccardo E. Marioni, Winfried Marz, Daniel L. McCartney, Karlijn A. C. Meeks, Lili Milani, Pashupati P. Mishra, Matthias Nauck, Ana Navas-Acien, Christoph Nowak, Annette Peters, Holger Prokisch, Bruce M. Psaty, Olli T. Raitakari, Scott M. Ratliff, Alex P. Reiner, Sylvia E. Rosas, Ben Schottker, Joel Schwartz, Sanaz Sedaghat, Jennifer A. Smith, Nona Sotoodehnia, Hannah R. Stocker, Silvia Stringhini, Johan Sundstrom, Brenton R. Swenson, Maria Tellez-Plaza, Joyce B. J. van Meurs, Jana V. van Vliet-Ostaptchouk, Andrea Venema, Niek Verweij, Rosie M. Walker, Matthias Wielscher, Juliane Winkelmann, Bruce H. R. Wolffenbuttel, Wei Zhao, Yinan Zheng, Marie Loh, Harold Snieder, Daniel Levy, Melanie Waldenberger, Katalin Susztak, Anna Kottgen and Alexander Teumer[#]. Meta-analyses identify DNA methylation associated with kidney function and damage, *Nature Communications*, 12, 7174. (2021)

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Software and Resources

- mV2G — A multiomic atlas for tissue-specific variant-to-gene prioritization https://mv2g.hbliulab.org
- Open4Gene — A R package for Peak-to-Gene linkage analysis in single cell multiome sequencing data https://github.com/hbliu/Open4Gene
- SMART — A Python package for deep analysis of DNA methylation by whole genome bisulfite sequencing http://fame.edbc.org/smart
- QDMR — A Java Package for identifying differentially methylated regions across large-scale datasets http://fame.edbc.org/qdmr
- QDCMR — A Java package for quantifying chromatin modification difference from ChIP-seq data https://github.com/hbliu/QDCMR
- SEA — A comprehensive online archive and analysis platform for super-enhancers for 11 species http://sea.edbc.org
- DiseaseMeth — A web-based resource and analysis platform for abnormal DNA methylation of human diseases http://diseasemeth.edbc.org
- MethyMarks — A web-based resource platform for tissue-specific methylation marks in human http://fame.edbc.org/methymark
- ScoreCard — eGFRcrea GWAS and genetic scorecard based on 2.2 million individuals https://susztaklab.com/GWAS2M
- Kidney_Epi_Pri — A pipeline for prioritization of disease-causal genes by integrating GWAS with the epigenome https://github.com/hbliu/Kidney_Epi_Pri
- eGFR_GWAS — eGFRcrea GWAS and Prioritization Atlas based on 1.5 million individuals https://susztaklab.com/GWAS
- Kidney_meQTL — Human kidney methylation QTL atlas based on 443 human kidneys https://susztaklab.com/Kidney_meQTL
- Kidney_eQTL — Human kidney expression QTL atlas based on 686 human kidneys https://susztaklab.com/Kidney_eQTL
- HKOCA — Human kidney open chromatin atlas based on 57,282 cells from 12 cell types of human kidneys https://susztaklab.com/Human_snATAC

Grants

- 2025 – 2030 — National Institute on Aging, #1P30AG092746-01 (PI: Vera Gorbunova)
- 2024 – 2029 — National Institute on Aging, #2P01AG047200-11 (PI: Vera Gorbunova)

Scientific Society Member

- 2026 – present — Membership of the Association of Chinese Americans in Cancer Research (ACACR)
- 2025 – present — Affiliate Membership of the Impact of Genomic Variation on Function Consortium (IGVF)
- 2022 – present — Membership of the American Society of Human Genetics (ASHG)
- 2019 – present — Membership of the American Society of Nephrology (ASN)
- 2018 – 2020 — Membership of the American Association for Cancer Research (AACR)

Honors and Awards

- 2026 — Young Investigator Travel Award of the ACACR 2026 Career Development Workshop
- 2025 — Moderator for the oral abstract session of the ASN Kidney Week 2025
- 2022 — Poster Prize in the Penn-Stanford CVI Symposium at the University of Pennsylvania
- 2019 — Poster Prize in the Epigenetics Symposium at the Franklin Institute
- 2017 — Science and Technology Award, Heilongjiang Province People's Government
- 2014 — National Scholarship for Doctoral Students, Ministry of Education of China
- 2012 — Science and Technology Award, Education Department of Heilongjiang Province

- 2011 Science and Technology Award, Science and Technology Department of Heilongjiang Province
- 2010 Outstanding Master's Degree Graduates, Heilongjiang Province Office of Education

Invited Talks

- 08/30/2026 *Invited speaker*, ACACR 2026 Career Development Workshop (Ann Arbor, MI; link)
- 11/25/2025 *Invited speaker*, The 4th Nutsheller Symposium (Virtual)
- 11/21/2025 *Invited speaker*, Center for Integrated Research Computing (CIRC) Symposium (Rochester, NY; link)
- 11/02/2025 *Invited speaker*, Precision Medicine and Intelligent Health Meeting, Fujian Medical University (Virtual)
- 10/29/2025 *Invited speaker*, Aab Cardiovascular Research Institute, University of Rochester (Rochester, NY)
- 10/24/2025 *Invited speaker*, Aging Research Day 2025, University of Rochester (Rochester, NY; link)
- 10/15/2025 *Invited speaker*, American Society of Human Genetics|ASHG 2025 Annual Meeting (Boston, MA; link)
- 09/17/2025 *Invited speaker*, Seminar visit, Department of Genetics, University of Georgia (Athens, GA; link)
- 08/08/2025 *Invited speaker*, ICIBM 2025, organized by the Ohio State University (Columbus, OH; link)
- 06/11/2025 *Invited speaker*, Summer Applied Genomics Seminar, University of Rochester (Rochester, NY; link)
- 04/08/2025 *Invited speaker*, Data Science in Medicine Seminar Series, Boston University (Virtual; link)
- 04/05/2025 *Invited speaker*, Chinese Genomics Meet-up (CGM) (Virtual; link)
- 03/31/2025 *Invited speaker*, The 3rd International Conference on Single-cell and Spatial Omics (Virtual; link)
- 03/28/2025 *Invited speaker*, The MCBIOS Annual Meeting 2025, University of Utah (Salt Lake City, UT; link)
- 03/20/2025 *Invited speaker*, Biomedical Genetics Department Seminar, University of Rochester (Rochester, NY)
- 02/05/2025 *Invited speaker*, Genetics, Epigenetics and Metabolism (GEM) Program Meeting (Rochester, NY; link)
- 01/03/2025 *Invited speaker*, Nephrology Academic Seminar, Peking Union Medical College Hospital (Virtual)
- 12/26/2024 *Invited speaker*, The 3rd Nutsheller Symposium (Virtual)
- 12/18/2024 *Invited speaker*, Chromatin Collective, University of Rochester (Rochester, NY)
- 11/27/2024 *Invited speaker*, Renal Clinical Research Seminar, University of Rochester (Rochester, NY)
- 06/26/2024 *Invited speaker*, Summer Applied Genomics Seminar, University of Rochester (Rochester, NY; link)
- 10/07/2022 *Lightning round presenter*, Annual MidAtlantic Bioinformatics Conference (Philadelphia, PA; link)
- 08/08/2022 *Invited speaker*, ICIBM 2022, organized by the Ohio State University (Philadelphia, PA; link)
- 05/19/2022 *Invited speaker*, Chinese Genomics Meet-up (CGM) (Virtual; link)
- 04/25/2022 *Invited speaker*, Penn Genetics 2022 Global Scientific Symposium (Philadelphia, PA)
- 04/14/2022 *Invited speaker*, University of Science and Technology of China (Hefei, China)
- 03/06/2019 *Invited speaker*, Keystone symposia: Unraveling the Secrets of Kidney Disease (BC, Canada)
- 07/10/2016 *Invited speaker*, The International Symposium on the Frontier of Big Data in Science (Baotou, China)

Poster Presentation

- 04/27/2026 *Poster presenter*, Nathan Shock Centers Annual Meeting (Bethesda, MD)
- 10/22/2025 *Poster presenter*, The IGVF Annual Meeting 2025, Durham Convention Center (Durham, NC)
- 05/13/2025 *Poster presenter*, The 35th Annual Genetics Day, University of Rochester (Rochester, NY)
- 09/18/2024 *Poster presenter*, Epigenetics & Chromatin (Cold Spring Harbor, NY)
- 05/13/2024 *Poster presenter*, The 35th Annual Genetics Day, University of Rochester (Rochester, NY)
- 10/24/2022 *Poster presenter and award*, The Penn-Stanford CVI Symposium (Philadelphia, PA)
- 06/02/2022 *Poster presenter*, Upenn Department of Medicine 2022 Research Day, (Philadelphia, PA)
- 06/02/2022 *Poster presenter*, HUP/CHOP Renal Research Symposium (Philadelphia, PA)
- 03/15/2022 *Poster presenter*, Penn IDOM Spring Symposium (Philadelphia, PA)
- 12/16/2019 *Poster presenter and award*, Fox Chase Cancer Center Epigenetics Symposium (Philadelphia, PA)
- 11/05/2019 *Poster presenter*, Kidney Week 2019, American Society of Nephrology (Washington DC)

- 03/19/2019 *Poster presenter*, Penn IDOM Spring Symposium (Philadelphia, PA)
- 12/17/2015 *Poster presenter*, 8th International Stem Cell and Regenerative Medicine Forum (Guangzhou, China)

Professional Service

Teaching

- 2026 – IND 432: Foundations of Modern Biology, Co-instructor (University of Rochester, Course link)
- 2025 – PTH 507: Cancer Biology, Co-instructor (University of Rochester, Course link)
- 2025 – PTH 510: Cell Biology of Disease, Co-instructor (University of Rochester, Course link)
- 2025 – IND 419: Introduction to Quantitative Biology, Co-instructor (University of Rochester, Course link)
- 2024 – IND 484: Current Topics in Bioinformatics, Co-instructor (University of Rochester, Course link)
- 2024 Biomedical Data Science Hackathon Summer 2024, Instructor (University of Rochester, Course link)
- 2015 – 2016 Computational Epigenetics, Co-instructor (Harbin Medical University)
- 2015 – 2016 Biomolecular Network Analysis, Instructor (Harbin Medical University)
- 2015 – 2016 Combinatorics and Graph Theory, Instructor (Harbin Medical University)
- 2011 – 2012 Database Principles and Applications, Instructor (Harbin Medical University)
- 2010 – 2011 System Biology, Co-Instructor (Harbin Medical University)

Postdoctoral Fellows Trained

- 2025 – present Haozhe Zou, Ph.D. (joint w/Gorbunova & Seluanov Lab)
- 2025 – present Yulong Tang, Ph.D.
- 2025 – present Shuangshuang Feng, Ph.D.
- 2024 – present Mengying Zhang, Ph.D.

Graduate Students Trained

- 2026 – present Dean Y. Zhang (M.D./ Ph.D., MSTP, University of Rochester)
- 2025 – present Zhicheng (Austin) Jin (joint w/Gorbunova & Seluanov Lab) (Ph.D., BGG, University of Rochester)

Rotation Graduate Students

- 2025 Spring Jacqueline Morin (BGG, University of Rochester)
- 2025 Spring Nidhi Shah (CMPP, University of Rochester)

Graduate Student Committees

• 2026	Yi Ding	Advisor: Yufang Bi, Ph.D.	Shanghai Jiao Tong University
• 2025	Hannah Swan	Advisor: Matthew N McCall, Ph.D.	University of Rochester
• 2025	Maxwell Zacher	Advisor: Vera Gorbunova, Ph.D.	University of Rochester
• 2025 – present	Xiaomi Liu	Advisor: Jennifer Brisson, Ph.D.	University of Rochester
• 2025 – present	Elijah Sterling	Advisor: Kaixiong Ye, Ph.D.	University of Georgia

Undergraduate Students Trained

- 2026 – present Avani Chitnis (Cell and Developmental Biology, University of Rochester)
- 2025 – present Adrienne Milhauser (Computational Biology, University of Rochester)
- 2024 – present Naidhruva Deb (Microbiology and Anthropology, University of Rochester)
- 2024 – present Jessica Chen (Cell & Developmental Biology, University of Rochester)
- 2024 – 2026 Sreejato Chatterjee (UR Computational Biology; Current: Postbaccalaureate IRTA Fellow at NIH)
- 2024 – 2025 Christina Wang (UR Cell & Developmental Biology; Current: Ph.D. student at Northwestern Uni.)

- 2024 – 2025 Zihang Yu (UR Biomedical Engineering; Current: graduate student at Columbia University)
- 2015 – 2016 Shanshan Zhang (HMU Bioinformatics; Current: Postdoctoral Associate in Kellis Lab at MIT)
- 2014 – 2015 Shipeng Shang (HMU Bioinformatics; Current: Assistant professor at Chongqing Medical University)
- 2013 – 2014 Yunzhen Wei (HMU Bioinformatics; Current: Assistant professor at Guangdong Medical University)
- 2012 – 2014 Rangfei Zhu (HMU Bioinformatics; Current: CEO of Hangzhou Mugu Technology Co., Ltd)

Conference Organization Committee

- 2026 Program Committee, International Conference on Intelligent Biology and Medicine (ICIBM 2026)
- 2025 Workshop Committee, International Conference on Intelligent Biology and Medicine (ICIBM 2025)
- 2024 Trainee Committee, International Conference on Intelligent Biology and Medicine (ICIBM 2024)

Grant Review

- 2026 *Ad-hoc* reviewer, NIH Genetics of Health and Disease Study Section (GHD, link)
- 2025 *Ad-hoc* reviewer, NASA Space Biology: Research Studies
- 2024 *Ad-hoc* reviewer, UK Research and Innovation of the United Kingdom

Ad-hoc Journal Reviewer

Aging
Biochemical Society Transactions
Bioinformatics
Biology Methods & Protocols
BMC Medicine
Briefings in Bioinformatics
Briefings in Functional Genomics
Cell
Cell & Bioscience
Cell Death and Disease
Cell Metabolism
Cell Proliferation
Circulation: Genomic and Precision Medicine
Computational and Structural Biotechnology Journal
Computers in Biology and Medicine
Current Bioinformatics
Current Genomics
Database
Diabetologia
eLife
Epigenetics & Chromatin
Epigenomics
Frontiers in Aging Neuroscience
Frontiers in Cell and Developmental Biology
Frontiers in Genetics
Frontiers in Oncology
Genes
Genome Medicine
Healthcare
Human Molecular Genetics
Journal for ImmunoTherapy of Cancer
Journal of Cancer
Journal of the American Heart Association
Journal of the American Society of Nephrology
Journal of Translational Medicine
Kidney International
Kidney International Reports
MedComm
Methods
Molecular Oncology
Molecular Psychiatry
Molecular Therapy - Nucleic Acids
Nature Communications
Nucleic Acids Research
Oncotarget
Phenomics
PLOS Computational Biology
PLOS Genetics
PLOS ONE
Quantitative Biology
Science Translational Medicine
Scientific Data
The International Journal of Molecular Sciences
The Journal of Clinical Endocrinology & Metabolism

Journal Editorial Board

- 2026 – present *Genome Medicine* (Editorial Board, link)
- 2016 – present *Briefings in Functional Genomics* (Editorial Board, link)
- 2018 – 2019 *Frontiers in Genetics* (Guest Associate Editor, link)